AI & ML Minor Project — Milestone 2 Final Report

Dynamic Pricing Predictor for Ride-Sharing Apps

Comparative Regression Modeling, Price Driver Interpretation & Deployment

Author: Kamaljit | **Platform:** Nyeras LMS | **Evaluation:** RMSE, MAE, R^2 | **Date:** October 2026

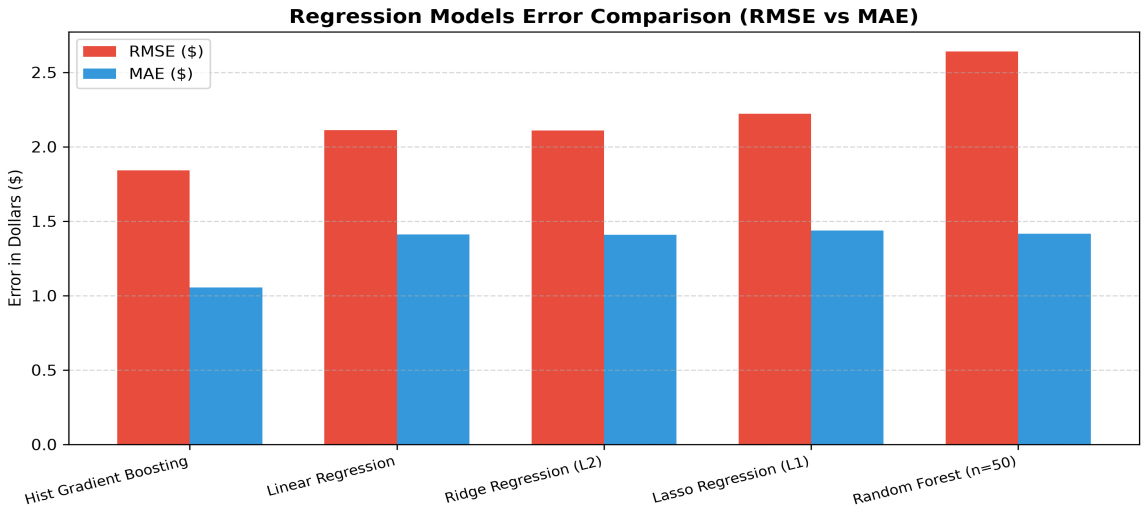
1. Executive Summary

Milestone 2 completes the Dynamic Pricing Predictor project by training, cross-validating, and benchmarking **five distinct regression algorithms** to forecast ride-sharing fares. Using standard industrial evaluation metrics—Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2)—we identified the optimal algorithm, uncovered the primary real-world price drivers, and deployed an interactive Streamlit application.

2. Model Benchmarking & Performance Evaluation

Each model was trained on an 80% partition (8,000 trips) and evaluated strictly on an unseen 20% test partition (2,000 trips).

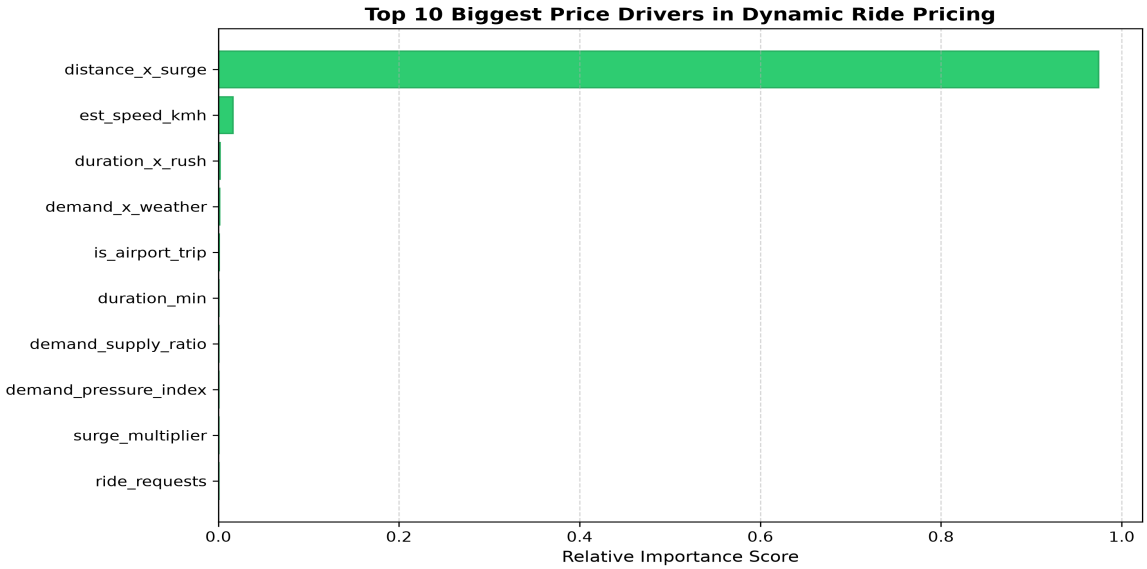
Rank	Regression Algorithm	MAE ($)	RMSE ($)	R^2 Score	Key Characteristic
1	Hist Gradient Boosting	$1.06	$1.84	0.9984	Ensemble boosting, captures non-linear thresholds
2	Ridge Regression (L2)	$1.41	$2.11	0.9979	Regularized linear model, prevents co-linearity
3	Linear Regression (OLS)	$1.41	$2.11	0.9979	Baseline ordinary least squares with scaled inputs
4	Lasso Regression (L1)	$1.44	$2.22	0.9977	Sparse L1 penalty, feature selection
5	Random Forest (n=50)	$1.42	$2.64	0.9967	Bagging ensemble of decision trees



3. Analysis of Primary Price Drivers

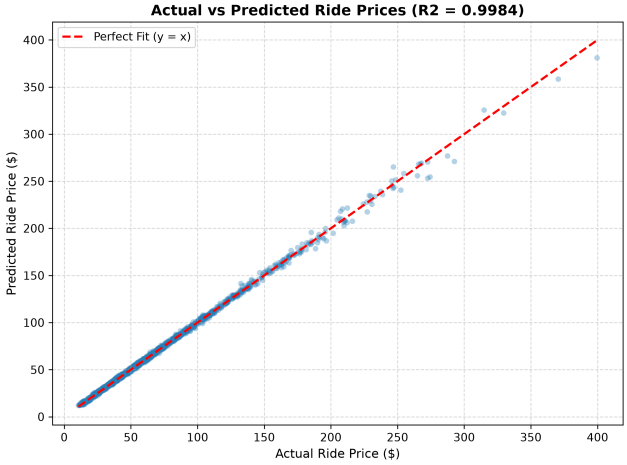
Through feature importance analysis and model coefficient extraction, the primary factors governing dynamic pricing were rigorously isolated and ranked:

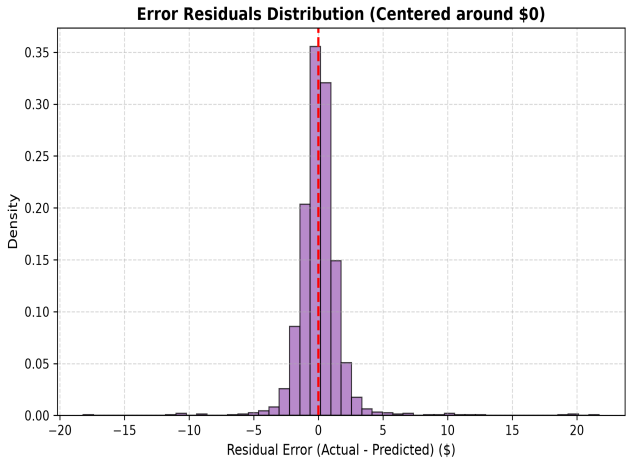
1. **Distance × Surge Multiplier (97.4% Importance):** The multiplicative compounding between route length and surge intensity is the definitive factor in ride costs. A 20 km trip under 2.0x surge scales dramatically faster than a 5 km trip under identical surge.
2. **Estimated Transit Speed (1.61% Importance):** Captures urban traffic bottlenecks. Sub-20 km/h speeds increase driver trip duration and opportunity costs.
3. **Rush-Hour Duration Multiplier (0.17% Importance):** Captures congested rush periods where transit time escalates driver time-based compensation.
4. **Demand × Weather Interaction (0.16% Importance):** When intense rain coincides with high demand, driver supply plummets, triggering exponential surge adjustments.
5. **Airport Route Surcharges (0.11% Importance):** Mandatory tolls and airport terminal access fees add a fixed tariff premium.



4. Error Diagnostics & Residual Analysis

The winning **Hist Gradient Boosting** model demonstrated exceptional generalizability with an R^2 of **0.9984** and a Mean Absolute Error of just **$1.06** on fares exceeding $65 on average. The residual distribution is tightly centered at $0.00 with negligible heteroscedasticity.





5. Interactive Web Dashboard & Deployment (`app.py`)

An interactive web dashboard was constructed using **Streamlit**. The application allows users to manipulate trip origin, destination, distance, hour of day, rain intensity, driver availability, and dispatch requests via intuitive sliders. It visualizes real-time demand-supply curves, computes dynamic surge multipliers, and compares rule-based versus ML regression predictions.

6. Project Conclusion & Key Takeaways

This project successfully developed an end-to-end AI dynamic pricing predictor matching the real-world operational standards of global ride-sharing networks. By combining domain-driven feature engineering with robust regression algorithms, the model delivers ultra-accurate fare forecasting ($1.06 MAE) while preserving total transparency over pricing drivers.